

14. $(\sqrt{3} + \sqrt{5})$ is one zero of a fourth-degree polynomial with integer coefficients and leading coefficient 1 when written in decreasing degree of x . Determine the constant term for this polynomial.
15. Determine the number of distinct arrangements of the letters in the word MATHEMATICS.
16. The roots for x in $x^3 + ax^2 + bx - 105 = 0$ are each two more than the roots for x in $x^3 - dx^2 + ex - f = 0$. a , b , d , e , and f are constant real numbers. Determine the value of the sum $(4d + 2e + f)$.
17. For all real values of θ such that $\cos(\theta) + \sin(\theta) \neq 0$, $\frac{\cos^3(\theta) + \sin^3(\theta)}{\cos(\theta) + \sin(\theta)} = k - w(\sin(2\theta))$. Determine the value of $(7k + 28w)$.
18. Let $t_1 = 77$ and $t_{n+1} = \frac{1 - 24t_n}{24 - 3t_n}$ for $n \geq 1$ be a sequence. Determine the value of t_{2016} . Express your answer as a common or improper fraction reduced to lowest terms.
19. Determine the number of points of intersection for the curves $y = 1 - \frac{x}{4}$ and $y = \frac{\sin x}{x}$ where x is measured in radians.
20. In the diagram with coordinates as shown, $AC = 51$ and $BC = 55$. The ratio of the area of $\triangle PAB$ to the area of $\triangle PAC$ to the area of $\triangle PBC$ is 1:2:3. Determine the **ordered pair** that represents point P. Express your answer as an **ordered pair** (k, w) with **each member** of the ordered pair expressed as a common or improper fraction reduced to lowest terms.

